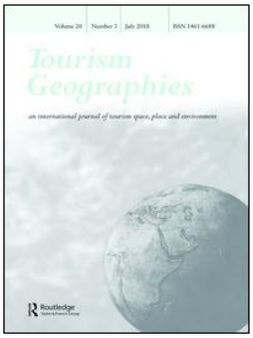

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EDITORIAL

Big data and tourism geographies – an emerging paradigm for future study?

Introduction

Big data has become a hot topic in tourism and geographic research, as well as in other socioeconomic sciences. The definition of big data remains open with characters of huge in volume, high in velocity, diverse in variety, exhaustive in scope, fine-grained in resolution, relational in nature, flexible and scalability (Kitchin, 2013). Geographers have noted that big data research has seen a shift from a data-scarce to a data-rich environment, resulting in the emergence of a new mode of the data-driven geography (Miller & Goodchild, 2015). In terms of big data analytical techniques, which include the ability to manage datasets that exceed the processing capability of traditional data analysis software, the phenomenon has almost become an all-purpose tool that often substitutes for traditional research approaches. At tourism industry conferences in China, big data companies demonstrate ever expanding technical abilities to monitor and manage the real-time visitor use at tourist attractions, and local government tourism administrators like to show off their cellphone screens with displays of these real-time monitoring tools for tourist sites under their governance. In turn, these officials are paying less attention to traditional data collection methods and models of tourist flows. Big data techniques have clearly brought opportunities and benefits for tourism research and industrial operations. However, does a cell phone display equal the big data? Will big data completely replace the theoretic models of tourist flows and other theories of tourism geographies? How will our understanding of tourism geographies change by big data? What aspects of tourism benefit the most from big data, and to what extent? These are questions needed to be examined properly, or at least thought about in a reasonable manner.

Big data and tourism geographic research issues

Geographical perspectives on the benefits of big data in tourism research mostly focus on the ease of new forms of data acquisition. Big data for tourism can be derived from a variety of sources (or channels), including cell phones, aerial imagery, satellite imagery, online social media platforms, merchandise sales and distribution sources, public monitoring devices (cameras and counters), and industry specific data collection services. New data channels emerge rapidly and provide large quantities of data that have potential applications for travel, tourism and hospitality research, and which can overcome the issues of data shortage that are common in traditional tourism research. More and more research has focused on the relationship between various big data indices and aspects of tourist flow, such as the Baidu index and tourist flow predictions (Huang, Zhang, & Ding, 2013; Li & Zhang, 2013; Sun, Zhang, Liu, & Zhang, 2017). Properly referenced big data can

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contribute rich information and insights into the patterns and structures of visitation and movement of visitors across geographic areas. A Chinese website company, for example, released a report on the stampede of December 31, 2014 in Shanghai soon after the tragedy based on big data analysis (Yang, 2015). The speed with which this report was issued made public aware that big data has the potential to manage and even prevent such disasters, although some researchers doubt if a credible forecast of that event could have been made based solely on big data.

The realm of the big data research, however, remains fluid and developing, because its definition is closely related to emerging technologies, new abilities in data processing, and innovations in applications. There are no clear thresholds of the absolute volume for datasets for them to be called 'big' data. Some data are bigger than others, and some applications require bigger data than others.

Cyberspace (the Internet) has received more attention than tourist tracking data, and is often described as a metaphor of the real world. The real world counterparts of cyberspace appear in tourism, such as theme parks that are based on virtual worlds created through online gaming and other entertainment media. Conceptually, it might be possible to think of all of tourism as a cyberworld metaphor from a modeling perspective. Software that simulates hotel and airline operations are examples of narrow applications of this type. Moreover, Liu et al. (2015) proposed the term 'social sensing' to describe the ways that big data is collected from various channels, from social media to satellite imagery, to monitor human social behavior.

The concept of big data is based largely on the internet technology, and e-commerce embodies the internet from a business perspective. Anderson (2006) proposed the concept of 'long-tail' e-commerce for customized niche marketing, which has also been applied to the tourism economy (Lew, 2008; Pan & Li, 2011). Using big data approaches, e-commerce can create large markets for niche products. According to the most popular legend, in 2009, students on the campus of Nanjing University in China complained about their single status by proclaiming the 'Singles Day' on 11.11 (Nov. 11th). This was picked up by the Ali-Baba company (China's leading online marketplace company), which turned the Singles Day into a shopping event. On November 11, 2017 the event generated total sales of 168,200 million RMB Yuan (about US$25.300 billion) for Ali-Baba, up from the 52 million RMB Yuan during the same day in 2009 (Xinhua net, 2017), i.e. sales multiply more than 3000 times after 8 years. Thus, a local tiny social event on a college campus was transferred into a global economic phenomenon, reaching consumers in over 200 countries and regions, through big data (in a broad sense) techniques and culture. Meanwhile, the event has also become a social-economic carnival show with live broadcasting by three satellite TV channels and many online broadcast websites, with live showings of peak moments of online sales, such as: 11 second to reached 100 million RMB Yuan, 3 minute reached 10 billion, 40 minute 50 billion, 9 hours reached 100 billion, and 24 hours to reach 168.2 billion (Sina Tech, 2017).

Characteristics commonly associated with big data, as noted above, include volume, variety, and velocity (Kitchin, 2013). Yet are other important features include value (with more valuable finer grain data), vertigo (data redundancy and nonstructural data with uncertainty), and volunteered (social media adding more information). All these big data 'V' characteristics are configuring a new map for tourism research (Table 1).

It is necessary to understand the mechanism and the types of big data for tourism geographic research, to dig or filter the tourism geographic information with big data, and even to create new categories of big data for various tourism geographies. The large

Table 1. Big data related tourism geographic research.

Type of research	Role of big data	Result of big data in research	Big data categories
Big data aided research	Data of traditional tourism conceptual models of analysis: big data with volume, variety, velocity, value, vertigo and volunteer	Expansion of traditional tourism research with more volume and variety of data	Data capturing and mining, filtering, storage, data cloud, analysis, search, sharing, transfer, visualization, querying, updating, social emotion, system robustness, information security, privacy and data source, big data platform for business operation
Big data aimed research	Special method for/of big data, big data itself as object, more sophisticate data system, post big data tech, A.I.	New methods, data mining and capturing, data filter, paradigm shift, system robustness,	
Big data arised phenomena research	New tourism phenomenon arised / driven by big data, e.g., new consumer behavior, new model of tourist decision making	To understand and predict new phenomena arised by big data, etc.	
Big data application research	New model of tourism operation, new tourism spatial model of organization economic value creation	New services and operation of tourism, Longtail market, B2C trade platform, etc.	

variety of research topics and fields related to big data in tourism geography emerges as follows:

1. Identification and description of big data phenomena: to find the critical influence of big data in tourism geographical research and to avoid the abuse and misuse of big data; big data-related socio-cultural phenomena, including how and to what extent big data changes socio-economic organizations related to tourism, and how traditional geographic topics and theories might shift with big data perspectives. For example, in China, there are increasing numbers of local command centers with huge display screens that offer live monitoring of tourist sites, which can also be accessed by cellphone by site administrators. In the Chinese cultural and institutional context, local administrators tend to use this form of big data in tourism management as a way of building reputation in their technological prowess, rather than as any real analytic and management practice tool.
2. Big data and allied technological services in tourism operations, marketplace and marketing: characteristics of mobile online services using big data, for example, accommodations, travel agents and transportation; implications of cellphone position tracing data; long tail and other theories to understand the big data tourism economy; big data supported visitor management and congestion monitoring; and new advertising strategies and techniques using big data mining and filtering for tourism.
3. Big data quality and ownership: The open-source and elicit mathematic descriptions of some popular big data indices issued by commercial companies are still lacking, raising issues as to whether a technical big data 'black box' will cause credibility problems; possible big data monopolies could be problematic for both researchers and industrial uses.
4. Social and technical accessibility to big data knowledge and applications: the technical knowledge gap between potential users and big data suppliers is increasing as new and more complex products are offered; attitudes, perceptions and abilities may vary

among different sectors of tourists and tourism providers toward big data and related info-tech service providers; large portion of the general public face user constraints to big data application, including infrastructure limitations, technical abilities, and psychological barriers. On the other hand, mobile online APPs (such as WeChat and Twitter) are also more accessible for larger numbers of users, and have become new resources or platforms for tourism operation.

Big data and tourism geography methodological and paradigm issues

Methodologically, big data techniques can help promote a more integrated approach to tourism management that encompasses a greater diversity of tourism sector activities, as well as its broader social and economic interconnections. Big data will expand and improve tourism research and by bringing new research topics and methodologies. These include:

1. Big data both as a technical tool to describe the tourism system and tourist flows with new sources of data, data support, and as tools for monitoring and research emerge. Big data mining through social media sensing, for example, has already been widely applied (Satta, Parola, Penco, & Persico, 2015). Big data often possess characteristics of nonstructured data, rather than traditionally structured data, which raises issues that have yet to be worked out.
2. The live (or near live) availability of some big data sources will require new mathematical models of tourism systems to match the fast data updates and feedbacks. Big data allows sophisticated description, such as tourists gathering in limited micro-spaces, the temporal pattern of visitor flows, and the possible application of short-term forecasting with trend extension hypotheses of various time series scales.
3. New measuring models and indices will be needed for emerging tourism statistics. Beyond simple trend extensions, the 'why and how' questions of tourism geography will become increasingly important for long-term forecasting with big data. Geographic understandings of spatial organization, spatial behavior and place theory can provide basic building blocks for the research, for example, on tourist crowding and congestion.

Big data might face a statistical paradox, since the basic of traditional statistics is to find the character of 'big' data object with small data sampling. The big size of a dataset is not necessarily the advantage of big data. Instead, the widening range of data types and channels for data collection may be its greatest advantage comparing to traditional statistics. In this sense, big data is not only a quantitative promotion, but also offers a qualitative shift (Barnes, 2013). In tourism research, big data techniques provide new and diverse data sources and channels, many of which may inform qualitative aspects of tourism. Social sensing (Liu et al. (2015), in particular, along with other big data techniques provide both quantitative and qualitative big data for tourism research, which might lead to changes in the basic view of the tourism system.

Ontologically, big data brings new tourism geographic phenomena, that is, the big data arised phenomena like newly operation tourism system based on big data. Moreover, big data largely changes human knowledge both in terms of our perception (and cognition) of the world, and in how we organize knowledge itself. Hence, big data becomes a paradigm issue. For example, big data now involves the daily life of the general public. Cellphone-based map applications are changing the human mental maps, spatial behavior and the model of people know and exist in the world. The basic function of mental maps for

guiding might not be needed when people have and ever-present big data supported cell-phone guidance system. Searching a map-related application, which utilizes big data and AI techniques, matches the user's need more efficiently, while at the same time creating knowledge restrictions. Such restrictions may be intentional user-created filters or filters applied by the app creator or app advertisers. Either way, the user is presented with a more narrowly spectrum of knowledge. The concept and cognition of space and place are different and changed by the context of big data, in comparison to direct (nondigitally mediated) bodily experiences.

Epistemologically, this new mode of understanding human behavior in a tourism world stretches traditional research assumptions in which description is a first step, followed by mechanism of causal relationships. However, under big data, the world is dominated by connective relationships, rather than causal relationships. This creates an image (or illusion?) that people who have big data can do anything in tourism research without relying on underlying structured principles or mechanisms. And just like a constructed descriptive model of gravity can be created by physicists for the precise quantitative relationship without understanding gravity's explicit mechanics, tourism geographers might only need to set up a connective relational model of the tourism system from the big data, and even build a framework of the tourism system from bottom up through big data and without references to traditional theoretical models. Such phenomenon goes beyond the traditional science paradigm that is emphasized in tourism and geographic research.

Concluding remarks

Table 1 suggests four 'A' types of big data research approaches in tourism: aided, aimed, arised, and application. New paradigm of big data research seems to be emerging, and tourism research is gradually shifting in response, from models consisting of phenomenon-mechanism-reality to those that consist of phenomenon-relationship-practical monitoring. Whether such a paradigm would be sophisticated, stable, and explicit, remains to be studied. Yet such a shift still needs to fulfill the basic component of a paradigm, i.e. the assumption and hypothesis of the view of the world (tourism system) and a related epistemology (theory and methodology). Whether the big data paradigm is confined to a narrowly technical context, somewhat like the chaos theory paradigm in the social sciences, or becomes a revolutionary and all pervasive approaches for tourism geographies, it will attract the interests of tourism geographers for at least the near future.

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